

From: Neil Nunn neil.j.nunn@gmail.com

Subject: Re: Jordan River progress

Date: October 10, 2023 at 4:00 PM

To: Michelle Unger Michelle.Unger@teck.com

Cc: Jerade, Liliana ENV:EX Liliana.Jerade@gov.bc.ca, dave burt dburt_and_assoc@telus.net, Rod Bealing rodbealing123@gmail.com, jonathan.croston Jonathan.Croston@snclavalin.com, Helen Jones research@pacheedaht.ca, Jennifer McConnachie Jennifer.McConnachie@teck.com, Wayne Jackaman w.jackaman1@gmail.com, Deborah Curran dlc@uvic.ca

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Hi Michelle, I hope you have had a good thanksgiving weekend. Thank-you for your update and taking the time to explain things on your end. It has taken me a little bit to reply as I wanted to review some of the past materials on the mine-waste contamination in the Jordan River prior to responding.

I have to be honest, I was a little surprised in your last communication to hear that Teck is considering dealing with the waste problem without physical remediation. When we met in 2021 and the plan for the interim cover was explained, Ken and Dave were quite open about their concerns that this interim solution would eventually become a permanent long term solution. You assured us that this would only be temporary measure until the proper course of action was determined.

In Q3 of this coming year it will be 10 years since Liliane and the BC Ministry of Environment sent an email to round table confirming that the site was high risk and asking members to take a "back seat and let the Ministry process continue" and to "trust" that the process in place will "help restore the habitat at the Jordan River."

It has been also almost 10 years since the initial letter was sent to Teck stating that high-risk site would "warrant involvement of the ministry to ensure appropriate and timely action takes place." It is our understanding that significant concentrations of copper still flow from the mine, through the waste pile, and directly into the river. And that during high flow releases from the dam, that the contaminated material is being eroded and settling in the lower stem.

I know that I am not alone in having concerns that in nearly ten years such little movement has been made to address the contamination, with worries about how much longer it be until the toxic impacts of this high-risk site are adequately remediated.

Michelle, thanks for you offer to meet anytime to discuss this matter. I would like to take you up on your offer to connect and discuss this further. I am wondering if we can find time in the next month to touch base. Anytime after the 16th Oct. works for me. If you were to throw some days out, I can see how it syncs with others' schedule who might want to be included in the call.

Thank-you,
Neil

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On Mon, Jun 5, 2023 at 11:49 AM Michelle Unger <Michelle.Unger@teck.com> wrote:

Hi Neil, nice to hear from you and hope you are enjoying the spring weather. We are unfortunately unable to come and provide an update in person due to other travel commitments however we can share that Teck is continuing to conduct studies at the Sunro site in 2023. Below is a summary of the main work that is scheduled to be completed in 2023.

- **Cover Performance Effectiveness Monitoring Program** - In 2022, subsequent to the installation of the interim cover over the North Waste Rock Pile in late 2021, Teck initiated an effectiveness monitoring program including sample collection and installation of conductivity probes in strategic locations. The objectives of this program are to i) measure water quality in the Jordan River downstream of the cover and 5100 Portal to document changes, ii) build upon the existing dataset to assess trends and support future remedial efforts at the site, and iii) assess the impact the cover has had on overall metal concentrations in the surface water of the Jordan River, downstream of the 5100 Portal. Specifically, the sample locations were selected from the baseline dataset (pre-liner locations) to determine if the seasonal increases in metals concentrations observed in porewater have been mitigated. To date, three effectiveness monitoring program sampling events have occurred and two more are planned, one for the summer 2023 and one for early winter of 2024. The results will be included in a standalone report expected to be submitted in Q2/2024.
- **Risk Assessment Objectives** - Teck will be retaining SNC Lavalin to conduct a human health and ecological risk opinion in the summer of 2023 to evaluate whether risks at the South Waste Rock Pile can be managed via institutional and engineered controls (i.e., without physical remediation). The risk opinion will quantitatively evaluate potential risk to human receptors of concern, including persons using the site for recreational purposes and Indigenous Peoples land users, and evaluate the potential for unacceptable risks to terrestrial ecological receptors at the community or population level. This evaluation will use the existing conceptual site model for the site, as well as data representing potential exposure point concentrations for both human and terrestrial ecological receptors of concern.

To support the risk opinion to date, a vegetation and wildlife survey was completed in 2021 to identify relevant terrestrial ecological receptors at the site and document the vegetation condition in the South Waste Rock Pile.

The results of the risk opinion will be summarized in a report expected to be submitted in Q2/2024. Teck anticipates that the risk opinion will support identification of possible mitigation measures that could be implemented to reduce exposures if deemed unacceptable. In the interim, activities completed at the site to limit human exposure to soil contaminants include gates and no trespassing signage on the access road.

Overall, Teck is continuing to use the ongoing studies and work done to date to evaluate what can be done to improve the site conditions with respect to both waste rock material and water management. Unfortunately, Teck also continues to grapple with technical challenges and access issues for this site and additional time is required to determine appropriate next steps. During this time, the Teck Legacy team expects to conduct a thorough review the conceptual site model following completion of the risk assessment and waste rock cover effectiveness programs described above and to develop a path forward for clarifying the existing uncertainties around the underground water balance. These exercises are expected to inform Teck's approach for the site and any additional studies that will be completed.

I hope this provides you and the JRRS with a suitable update.

Please feel free to contact me if you have any further questions or would like to meet to further discuss next steps.

Kind regards,

Michelle

From: Neil Nunn <neil.j.nunn@gmail.com>

Sent: Monday, May 8, 2023 2:12 PM

To: Michelle Unger <Michelle.Unger@teck.com>

Cc: Jerade, Liliana ENV:EX <Liliana.Jerade@gov.bc.ca>; dave burt <dburt_and_assoc@telus.net>; wjackaman@shaw.ca; Rod Bealing <rodbealing123@gmail.com>; jonathan.croston <Jonathan.Croston@snclavalin.com>; Helen Jones <research@pacheedaht.ca>; Jennifer McConnachie <Jennifer.McConnachie@teck.com>; Calvin Sandborn <csandbor@uvic.ca>

Subject: Re: Jordan River progress

[External email]

Hi Michelle, I hope this email finds you well. I am not doing the best job at filling Ken's shoes as the JR remediation "update reminder person," but better late than never!

Just checking in to see if there are still plans for Teck to update the JRRS this spring.

Best wishes,

Neil

On Wed, Dec 14, 2022 at 1:09 PM Michelle Unger <Michelle.Unger@teck.com> wrote:

Hi Neil, nice to hear from you and it's hard to believe that another year has gone by so

quickly. We have been in a year of growth within our Legacy Teck team and have a few new team members that will be starting to support the long term remediation plan for the Sunro site. We are definitely long overdue to meet with the JRRS and will be ready to set up a meeting in the spring of 2023 to discuss findings and next steps in planning.

Wishing you a happy and healthy holiday season.

Kind regards,

Michelle

From: Neil Nunn <neil.j.nunn@gmail.com>
Sent: Tuesday, November 29, 2022 10:06 AM
To: Michelle Unger <Michelle.Unger@teck.com>
Cc: Jerade, Liliana ENV:EX <Liliana.Jerade@gov.bc.ca>; dave burt <dburt_and_assoc@telus.net>; wjackaman@shaw.ca; Rod Bealing <rodbealing123@gmail.com>; jonathan.croston <Jonathan.Croston@snclavalin.com>; Helen Jones <research@pacheedaht.ca>
Subject: Re: Jordan River progress

[External email]

Hi Michelle,

I hope this finds you well. I was wondering if there are plans to share an update on how the temporary cover has been and if there has been any further consideration of permanent remediation plans?

My best wishes,

Neil

On Fri, Jul 23, 2021 at 7:30 AM Michelle Unger <Michelle.Unger@teck.com> wrote:

Hi Ken, it's hard to believe it's almost the end of July and I apologize for not getting back to you sooner with an update. I am not sure if you are aware but there was a change of land ownership of the property at the end of June from Western Forest Products to Forebay Holdings Ltd. I have included Rod Bealing as his is the contact for Forebay

Holdings.

Due to summer vacations I'm not sure when we can commit to a virtual meeting, but I will see if we can arrange a time mid August.

In the interim I have provided you with a copy of the semi-annual report that was submitted to Ministry of Environment Land Remediation group last week which outlines the activities conducted in Q1/Q2 2021 and what is planned for Q3/Q4 2021. You will see that Teck is proceeding with an interim cover to help mitigate contaminate loading and high-risk conditions in the river in the short term (i.e., while we determine the best long term approach for the Site). SNC evaluated cover options to reduce net infiltration rates at the North Waste Rock pile (NWRP). A number of factors were considered and ranked, and based on the cover rating matrix, the 40-mil textured, 16 oz net-weight geotextile, linear low-density polyethylene (LLDPE) geomembrane liner option with a Wind Defender was recommended to be installed over acid generating portions of the NWRP. Two Google Earth plan views of the approximate extent of the cover and image of the waste rock pile are provided below. We are currently finalizing the engineered cover design and are hoping to begin construction September 1st and to be completed by September 15th to comply with the safe work window for fish. As part of the remedial planning process we submitted notifications to DFO (Projects Near Water) and FLNRO (Section 11 Approval For Instream Work). Teck is also engaging with the Ministry of Mines to determine if a permit is required to conduct this work.

I am leaving for vacation today and will look into meeting dates when I return in August.

Kind regards,

Michelle





-----Original Message-----

From: KEN FARQUHARSON <ken.farquharson@shaw.ca>

Sent: Monday, July 19, 2021 11:02 AM

To: Michelle Unger <Michelle.Unger@teck.com>

Cc: Jerade, Liliانا ENV:EX <Liliana.Jerade@gov.bc.ca>; dave burt
<dburt_and_assoc@telus.net>; wjackaman@shaw.ca; neil.j.nunn@gmail.com

Subject: Jordan River progress

[External email]

Hi Michelle,

We were in contact about a month ago when I inquired as to when you would be able to offer a progress report. Please advise how the work is going and when you might be able to meet with local reps of the Round Table, Zoom or in person. Ken

Ken Farquharson

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